

achieve near-double-digit growth in the coming years, driven by rising demand for electronics-enabled safety and comfort features (Automotive Conformal Coatings Market, 2021). Modern vehicles are equipped with sensors supporting in-vehicle air quality systems, automatic braking systems (ABS), airbag actuation, and HVAC control. ECUs manage electrical sub-systems such as passenger seating, valve timing, anti-theft systems, and braking functions. Printed circuit boards (PCBs) enable the high-precision signal transmission required for engine, braking, and safety systems.

Overall, product selection involves multiple variables, with environmental exposure being the most critical.

Inside the vehicle cabin, where exposure to harsh chemicals is minimal, thermoplastic acrylic coatings and silicone-acrylic hybrid coatings are well-suited. They provide effective protection against moisture and dust and can be removed easily for rework if components are damaged.

Thermoset coatings offer excellent solvent resistance and can withstand exposure to road salt, fuel vapours, oils, and glycol ethers, making them ideal for ECUs.

A secondary consideration when choosing conformal coatings is cost; however, engineers and buyers tend to make the mistake of extrapolating price based on the bulk cost of the material (cost per litre, gallon, and so on). This analysis is incorrect because it fails to account for the coverage constraints of different coating options; it is best to evaluate the price per part at a given dry film thickness.

The choice of conformal coating considers many factors and is not always a straightforward, linear process. The strengths and weaknesses of each coating type must be weighed against the project's requirements and constraints. Fig. 2

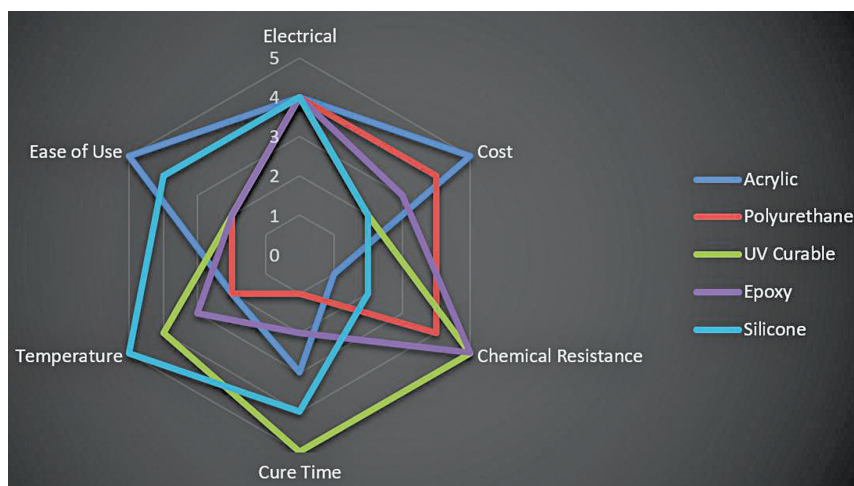


Fig. 2: Radar diagram comparing acrylic, polyurethane, UV cure, epoxy, and silicone conformal coatings

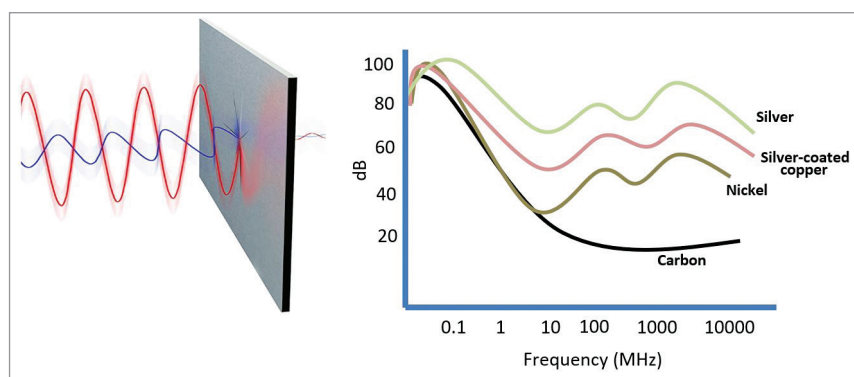


Fig. 3: Graphical representation of an attenuated signal passing through a conductive medium (left). On the right is a chart showing the relative shielding effectiveness of different conductive filler systems

is a summary chart comparing the main conformal chemistries.

EMP protection for vehicles

With increasing reliance on electronic sensors and controllers, electromagnetic interference presents a significant risk to automotive functionality. Invisible electromagnetic waves can disrupt circuit performance, much like crosstalk in electrical cables.

Conformal coatings protect ECUs, PCBs, and sensors from environmental exposure. In contrast, electrically conductive coatings shield components from interference caused by neighbouring electronics. The same criteria for selecting the right conformal coating apply to conductive coating chemistry, where acrylic and water-based coatings are appropriate for vehicle cabin

components. Conversely, thermosets like epoxy conductive coatings withstand chemical exposure well and are more suitable for sensors and components under the vehicle's hood.

Noise reduction, or attenuation, of incoming electromagnetic signals is another crucial consideration when choosing a coating, as it indicates the level of protection the device receives from the coating. Shielding effectiveness is a measure of the relative loss of power from a signal at a given frequency as it moves from a transmitter to a receiver through a medium. Shielding effectiveness is measured in decibels (dB), with higher values indicating greater protection. Fig. 3 shows the basic concept of attenuation and compares the relative shielding effectiveness of common